

Geometric Quantities for the FLRW Metric

Generated by EinsteinKPy2Tex

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1 Citation Information

If you use these results and code in your research, please cite:

- Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 844-847.
- Ribeiro, S., Bapat, A., Tandon, A., & EinsteinPy Developers (2020). EinsteinPy: A Python Package for General Relativity. *Journal of Open Source Software*, 5(51), 2356. <https://doi.org/10.21105/joss.02356>
- P. Ntelis (2026). EinsteinKPy2Tex: FLRW Metric Tensor Calculator (Version 1.0). <https://github.com/lontelis/EinsteinKPy2Tex>

2 Geometric quantities for the FLRW metric

The metric is given by

$$ds^2 = -c^2 dt^2 + a(t)^2(dx^2 + dy^2 + dz^2). \quad (1)$$

2.1 Metric tensor $g_{\mu\nu}$

$$g_{\mu\nu} = \begin{bmatrix} -c^2 & 0 & 0 & 0 \\ 0 & a^2(t) & 0 & 0 \\ 0 & 0 & a^2(t) & 0 \\ 0 & 0 & 0 & a^2(t) \end{bmatrix} \quad (2)$$

2.2 Christoffel symbols $\Gamma_{\nu\rho}^{\mu}$ (non-zero)

$$\Gamma_{xx}^t = \frac{a(t) \frac{d}{dt} a(t)}{c^2} \quad (3)$$

$$\Gamma_{yy}^t = \frac{a(t) \frac{d}{dt} a(t)}{c^2} \quad (4)$$

$$\Gamma_{zz}^t = \frac{a(t) \frac{d}{dt} a(t)}{c^2} \quad (5)$$

$$\Gamma_{tx}^x = \frac{\frac{d}{dt} a(t)}{a(t)} \quad (6)$$

$$\Gamma_{xt}^x = \frac{\frac{d}{dt} a(t)}{a(t)} \quad (7)$$

$$\Gamma_{ty}^y = \frac{\frac{d}{dt} a(t)}{a(t)} \quad (8)$$

$$\Gamma_{yt}^y = \frac{\frac{d}{dt} a(t)}{a(t)} \quad (9)$$

$$\Gamma_{tz}^z = \frac{\frac{d}{dt} a(t)}{a(t)} \quad (10)$$

$$\Gamma_{zt}^z = \frac{\frac{d}{dt} a(t)}{a(t)} \quad (11)$$

$$(12)$$

2.3 Riemann curvature tensor $R^\mu_{\nu\rho\sigma}$ (non-zero)

$$R^t_{xtx} = \frac{a(t) \frac{d^2}{dt^2} a(t)}{c^2} \quad (13)$$

$$R^t_{xxt} = -\frac{a(t) \frac{d^2}{dt^2} a(t)}{c^2} \quad (14)$$

$$R^t_{yty} = \frac{a(t) \frac{d^2}{dt^2} a(t)}{c^2} \quad (15)$$

$$R^t_{yyt} = -\frac{a(t) \frac{d^2}{dt^2} a(t)}{c^2} \quad (16)$$

$$R^t_{ztz} = \frac{a(t) \frac{d^2}{dt^2} a(t)}{c^2} \quad (17)$$

$$R^t_{zzt} = -\frac{a(t) \frac{d^2}{dt^2} a(t)}{c^2} \quad (18)$$

$$R^x_{ttx} = \frac{\frac{d^2}{dt^2} a(t)}{a(t)} \quad (19)$$

$$R^x_{txt} = -\frac{\frac{d^2}{dt^2} a(t)}{a(t)} \quad (20)$$

$$R^x_{yxy} = \frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (21)$$

$$R^x_{yyx} = -\frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (22)$$

$$R^x_{zxx} = \frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (23)$$

$$R^x_{zzx} = -\frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (24)$$

$$R^y_{tty} = \frac{\frac{d^2}{dt^2} a(t)}{a(t)} \quad (25)$$

$$R^y_{tyt} = -\frac{\frac{d^2}{dt^2} a(t)}{a(t)} \quad (26)$$

$$R^y_{xxy} = -\frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (27)$$

$$R^y_{xyx} = \frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (28)$$

$$R^y_{zyz} = \frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (29)$$

$$R^y_{zzz} = -\frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (30)$$

$$R^z_{ttz} = \frac{\frac{d^2}{dt^2} a(t)}{a(t)} \quad (31)$$

$$R^z_{tzt} = -\frac{\frac{d^2}{dt^2} a(t)}{a(t)} \quad (32)$$

$$R^z_{xxz} = -\frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (33)$$

$$R^z_{xzx} = \frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (34)$$

$$R^z_{yyz} = -\frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (35)$$

$$R^z_{zyz} = \frac{\left(\frac{d}{dt} a(t)\right)^2}{c^2} \quad (36)$$

2.4 Ricci tensor $R_{\mu\nu}$ (non-zero)

$$R_{tt} = -\frac{3\frac{d^2}{dt^2}a(t)}{a(t)} \quad (38)$$

$$R_{xx} = \frac{a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2}{c^2} \quad (39)$$

$$R_{yy} = \frac{a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2}{c^2} \quad (40)$$

$$R_{zz} = \frac{a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2}{c^2} \quad (41)$$

$$(42)$$

2.5 Ricci scalar R

$$\frac{6\left(a(t)\frac{d^2}{dt^2}a(t) + \left(\frac{d}{dt}a(t)\right)^2\right)}{c^2a^2(t)} \quad (43)$$

2.6 Einstein tensor $G_{\mu\nu}$ (non-zero)

$$G_{tt} = \frac{3.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \quad (44)$$

$$G_{xx} = \frac{-2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{c^2} \quad (45)$$

$$G_{yy} = \frac{-2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{c^2} \quad (46)$$

$$G_{zz} = \frac{-2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{c^2} \quad (47)$$

$$(48)$$

2.7 Geodesic equations

The geodesic equations are given by:

$$\ddot{t} = -\frac{\dot{x}^2 a(t) \frac{d}{dt} a(t)}{c^2} - \frac{\dot{y}^2 a(t) \frac{d}{dt} a(t)}{c^2} - \frac{\dot{z}^2 a(t) \frac{d}{dt} a(t)}{c^2} \quad (49)$$

$$\ddot{x} = -\frac{2\dot{t}\dot{x} \frac{d}{dt} a(t)}{a(t)} \quad (50)$$

$$\ddot{y} = -\frac{2\dot{t}\dot{y} \frac{d}{dt} a(t)}{a(t)} \quad (51)$$

$$\ddot{z} = -\frac{2\dot{t}\dot{z} \frac{d}{dt} a(t)}{a(t)} \quad (52)$$

$$(53)$$

where λ is an affine parameter and $\dot{x}^\mu = dx^\mu/d\lambda$.

2.8 Kretschmann scalar

The Kretschmann scalar $K = R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$ is:

Full expression (unsimplified)

$$\frac{12 \left(\frac{d^2}{dt^2} a(t) \right)^2}{c^4 a^2(t)} + \frac{12 \left(\frac{d}{dt} a(t) \right)^4}{c^4 a^4(t)} \quad (54)$$

2.8.1 Simplified expression

$$K = \frac{12 \left(a^2(t) \left(\frac{d^2}{dt^2} a(t) \right)^2 + \left(\frac{d}{dt} a(t) \right)^4 \right)}{c^4 a^4(t)} \quad (55)$$